

ArogyaMitr: Dual-Outlet Rural Healthcare Infrastructure & Voice-Guided Tele-Care Platform

Problem Statement ID: 26133 | Organization: Government of Maharashtra (MSInS) | Theme: MedTech / BioTech / HealthTech

Executive Summary

ArogyaMitr is a decentralized rural healthcare platform combining a physical dual-outlet infrastructure with a voice-guided, illiterate-friendly WebApp to bridge last-mile healthcare access. Our dual-outlet model establishes an **Inner-Village Care Point** for patient queries, basic triage, and assisted check-ins, supported by a shared **Outskirts Cluster Hub** that manages cold-chain medicine stock, diagnostic tests, and emergency logistics for 3–5 villages. The multilingual WebApp uses pictorial icons and regional voice prompts (Marathi/Hindi) for non-literate users. It enables direct doctor tele-consultations, instant prescription routing to the outskirts hub for rapid dispensing, 1-on-1 live human agent guidance, and one-touch 108 emergency SOS dispatch.

1. Problem Analysis & Rural Ground Realities

In rural and tribal regions of Maharashtra (including Gadchiroli, Melghat, and Nandurbar), healthcare delivery faces structural breakdowns across three dimensions:

- **The Digital Literacy Divide:** Over 45% of rural villagers cannot read or write complex English text or navigate standard telemedicine applications.
- **Travel Burden for Diagnostics & Drugs:** Even when villagers consult a doctor, they must travel 40–80 km to town centers simply to collect prescribed medicines and blood reports.
- **Emergency Isolation:** In acute emergencies (snakebites, cardiac distress, maternal complications), villagers lack one-touch panic connectivity or 1-on-1 human guidance to direct ambulances.

2. Proposed Solution Architecture

Component	Physical & Digital Architecture	Key Functions & Village Impact
Outlet 1: Gram Care Point <i>(Inside Village Core)</i>	Located at Gram Panchayat / Community Hall, equipped with a tablet kiosk and assisted by a local health volunteer.	Handles daily patient queries, preliminary triage, basic vitals screening (BP, SpO2), and assisted digital onboarding.
Outlet 2: Diagnostic & Supply Hub <i>(Outskirts Cluster Junction)</i>	Strategic highway facility serving a 3–5 village cluster (5–7 km radius), operating as a quick-commerce health micro-depot.	Houses cold-chain medicines, processes rapid diagnostic tests, dispenses lab reports, and stages 108 emergency ambulances.
Universal WebApp <i>(Illiterate-Friendly UX)</i>	Progressive Web App (PWA) with 100% pictorial icons, audio prompts in Marathi, Gondi, and Hindi, and zero-typing intake.	Enables non-literate and elderly citizens to speak or tap large visual symbols (Heart, Head, Pregnancy, Emergency).
1-on-1 Human Guide & SOS <i>(Direct Tele-Care)</i>	Live WebRTC video/audio channel to trained agents + One-Touch 108 Emergency SOS with GPS coordinate broadcasting.	Provides reassuring, live step-by-step guidance in native dialects and dispatches nearby ambulances under 15 minutes.

3. Technology Stack & Interoperability

- **Frontend & Voice Engine:** React / Flutter PWA with Bhashini Speech-to-Text & Text-to-Speech APIs for natural spoken interactions in Marathi and Hindi. High-contrast icon layout for accessibility.
- **Telehealth & 1-on-1 Guidance:** LiveKit WebRTC media engine dynamically throttling audio down to 15 kbps for seamless calls over weak 2G/3G networks.
- **Micro-Fulfillment & Inventory Ledger:** FastAPI (Python) backend with PostgreSQL 16 tracking real-time pharmaceutical inventory and diagnostic test orders between the Outskirts Hub and District Warehouses.
- **National Digital Health Integration:** Compliant with Ayushman Bharat Digital Mission (ABDM) ABHA ID linkage and HL7 / FHIR R4 clinical JSON schemas; compliant with the DPDP Act 2023.

4. Phased Implementation Roadmap

Phase	Timeline	Target Scope & Key Deliverables
Phase 1: Pilot Cluster	Months 1 – 3	Deploy in 10 Villages (Gadchiroli/Melghat) : 10 Inner Village Care Points + 2 Outskirts Diagnostic & Supply Hubs. Onboard 30 village health agents.
Phase 2: District Scale	Months 4 – 8	Scale across 5 High-Priority Rural Districts with 100 Outskirts Logistics Hubs serving 400+ villages; integrate with 108 Emergency Medical Services.
Phase 3: Statewide Rollout	Months 9 – 12+	Statewide deployment across all 36 districts of Maharashtra , federating with the Ayushman Bharat Digital Mission (ABDM) and national supply chains.

5. Measurable Impact & Competitive Edge

- **70% Reduction in Diagnostic Travel:** Eliminates long bus journeys to city centers for basic diagnostic tests and pharmaceutical pickup.
- **100% Inclusion for Non-Literate Citizens:** Voice-first prompts and large pictorial cues ensure no villager is excluded due to illiteracy.
- **< 15 Minute Emergency Response:** Direct 1-touch SOS coordinates immediate ambulance dispatch from the Outskirts Hub staging area.
- **Prescription-to-Dispense Continuity:** Tele-doctor prescriptions are auto-routed to the Outskirts Supply Hub for immediate preparation.